

BACK PROPOGATION

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[9] 25s ▶ # Step 1: Import required libraries
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Input

# Step 2: Create a small dataset
# [Study Hours, Attendance]
X = np.array([
    [1, 40], [2, 50], [3, 55], [4, 65],
    [5, 70], [6, 75], [7, 80], [8, 90]
], dtype=float)

# 0 = Fail, 1 = Pass
y = np.array([0, 0, 0, 0, 1, 1, 1, 1])

# Step 3: Normalize the input values
X_norm = X.copy()
X_norm[:, 0] = X_norm[:, 0] / 8      # Normalize study hours
X_norm[:, 1] = X_norm[:, 1] / 100    # Normalize attendance

# Step 4: Build the neural network
model = Sequential([
    Input(shape=(2,)),
    Dense(4, activation="relu"),
    Dense(1, activation="sigmoid")
])

# Step 5: Configure the model
model.compile(
    optimizer="adam",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
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[9] metrics=["accuracy"]
)

# Step 6: Train the model
print("Training model...")
model.fit(X_norm, y, epochs=500, verbose=0)
print("Model training completed!\n")

# Step 7 & 8: Test and display at least five student outputs
test_students = [
    {"hours": 2, "attendance": 45}, # Expected Fail
    {"hours": 4, "attendance": 60}, # Borderline
    {"hours": 6, "attendance": 80}, # Expected Pass
    {"hours": 8, "attendance": 95}, # Strong Pass
    {"hours": 1, "attendance": 90} # Low study, high attendance
]

print("--- PREDICTION RESULTS ---")
for student in test_students:
    h, a = student["hours"], student["attendance"]
    norm_input = np.array([[h / 8, a / 100]])
    prob = model.predict(norm_input, verbose=0)[0][0]
    status = "PASS ✅" if prob >= 0.5 else "FAIL ❌"
    print(f"Input: {h}h Study, {a}% Attendance | Confidence: {prob:.2%} | Result: {status}")
```

OUTPUT:

```
... Training model...
Model training completed!

--- PREDICTION RESULTS ---
Input: 2h Study, 45% Attendance | Confidence: 44.05% | Result: FAIL ❌
Input: 4h Study, 60% Attendance | Confidence: 47.92% | Result: FAIL ❌
Input: 6h Study, 80% Attendance | Confidence: 51.44% | Result: PASS ✅
Input: 8h Study, 95% Attendance | Confidence: 53.95% | Result: PASS ✅
Input: 1h Study, 90% Attendance | Confidence: 38.31% | Result: FAIL ❌
```